

Perimeter and Area of Coordinate Polygons

Answer Key

High School Geometry Coordinate Geometry

Quick Answers

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|-----------|---------------------------|
| 1. 10, 24 | 5. 0, 0, 0, 0, 5, 20 |
| 2. 40 | 6. $-\frac{1}{2}$, 2, 20 |
| 3. 5, 22 | 7. 64 |
| 4. 25 | 8. $\sqrt{29}$, 29 |
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Curriculum

Level: High School Geometry Coordinate Geometry

HSG-GPE.B.4 – *problems 5, 8*

HSG-GPE.B.5 – *problem 6*

HSG-GPE.B.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–4 of 5 across 17 tagged checks.

1. Sail $A(2, 1)$, $B(10, 1)$, $C(10, 7)$ — seam length and perimeter.

Step 1. AB is horizontal and BC is vertical, so those two lengths are read straight off the coordinates: $AB = 10 - 2 = 8$ and $BC = 7 - 1 = 6$. Only AC needs the distance formula.

Step 2. $AC = \sqrt{(10 - 2)^2 + (7 - 1)^2} = \sqrt{64 + 36} = \sqrt{100}$, so the slanted seam is 10

Step 3. The perimeter adds the three seams: $8 + 6 + 10$, so the sail takes 24 m of seaming.

Common wrong answer: 14 for AC , from adding the two legs. A slanted segment is always shorter than the two legs it spans — that is the triangle inequality, and it is a fast sanity check.

2. Slab $(-3, -2)$, $(5, -2)$, $(5, 3)$, $(-3, 3)$ — area.

Step 1. Opposite sides share an x - or a y -value, so the rectangle is axis-aligned and its side lengths are coordinate differences.

Step 2. Width: from $x = -3$ to $x = 5$ is 8. Height: from $y = -2$ to $y = 3$ is 5.

Step 3. Area is width times height, 8×5 , so the slab covers 40 m²

Common wrong answer: 26, which is the perimeter $2(8) + 2(5)$. Perimeter is a length in metres; area is in square metres. Checking the unit catches this before the number does.

3. Tile $P(0, 0)$, $Q(6, 0)$, $R(9, 4)$, $S(3, 4)$ — **perimeter.**

Step 1. PQ lies on the x -axis, so $PQ = 6$. The slanted side QR needs the distance formula, and because $PQRS$ is a parallelogram the opposite sides repeat those two lengths.

Step 2. $QR = \sqrt{(9-6)^2 + (4-0)^2} = \sqrt{9+16} = \sqrt{25}$, so $QR = \boxed{5}$

Step 3. Perimeter $= 2(6) + 2(5)$, so the tile measures around.

22 units

Teaching note: confirm the parallelogram before halving the work — $\overrightarrow{PQ}$ and $\overrightarrow{SR}$ are both $(6, 0)$, which is what makes the two long sides equal.

4. Flowerbed $(0, 0)$, $(8, 2)$, $(3, 7)$ — **area.**

Step 1. No side is horizontal or vertical, so there is no base to read off the grid. The shoelace formula takes the vertices in order and needs no side lengths at all.

Step 2. Taking the vertices counter-clockwise:

$$A = \frac{1}{2} |0(2-7) + 8(7-0) + 3(0-2)| = \frac{1}{2} |0 + 56 - 6| = \frac{1}{2}(50)$$

Step 3. The flowerbed covers

25 square units

Common wrong answer: 56, the area of the smallest rectangle around the triangle. A triangle inside a rectangle never fills more than half of it, so an answer above 28 here cannot be right.

5. Banner $A(-4, 0)$, $B(0, 3)$, $C(4, 0)$, $D(0, -3)$ — **diagonals and perimeter.**

Step 1. A midpoint is the average of the endpoints, taken one coordinate at a time. Diagonal AC runs from $(-4, 0)$ to $(4, 0)$ and diagonal BD from $(0, 3)$ to $(0, -3)$.

Step 2. Midpoint of AC : $\left(\frac{-4+4}{2}, \frac{0+0}{2}\right) = \boxed{(0, 0)}$

Midpoint of BD : $\left(\frac{0+0}{2}, \frac{3+(-3)}{2}\right) = \boxed{(0, 0)}$

The two diagonals share a midpoint, so each one cuts the other exactly in half.

Step 3. Side $AB = \sqrt{(0-(-4))^2 + (3-0)^2} = \sqrt{16+9} = \sqrt{25}$, so $AB = \boxed{5}$. With all four sides equal the perimeter is $4(5)$, or

20 units

Teaching note: equal diagonal halves plus perpendicular diagonals is the coordinate signature of a rhombus, and this banner is one — which is exactly what licenses multiplying a single side by four.

6. Park $L(1, 2)$, $M(5, 0)$, $N(9, 8)$ — **slopes and area.**

Step 1. Slope compares rise to run, so it is the tool for deciding whether the corner at M is a right angle. Compute both slopes that meet there.

Step 2. $\text{slope}(LM) = \frac{0-2}{5-1} = \boxed{-\frac{1}{2}}$

$\text{slope}(MN) = \frac{8-0}{9-5} = \boxed{2}$

Their product is -1 , so $LM \perp MN$ and the corner at M is a right angle.

Step 3. With a right angle at M the two sides meeting there are the base and the height: $LM = \sqrt{16+4} = 2\sqrt{5}$ and $MN = \sqrt{16+64} = 4\sqrt{5}$, so $A = \frac{1}{2}(2\sqrt{5})(4\sqrt{5}) = \frac{1}{2}(40)$. The park covers

20 square units

Check it the other way: shoelace on the same three vertices gives $\frac{1}{2}|1(0-8) + 5(8-2) + 9(2-0)| = \frac{1}{2}(40)$, the same number without needing the right angle at all.

7. Building lot $(0, 0)$, $(10, 0)$, $(10, 6)$, $(2, 8)$ — area.

Step 1. Two sides are axis-aligned and two are slanted, so there is no single length-times-width shortcut. Shoelace handles the whole quadrilateral in one pass, with the vertices taken in order around the boundary.

Step 2.

$$A = \frac{1}{2} |0(0 - 8) + 10(6 - 0) + 10(8 - 0) + 2(0 - 6)| = \frac{1}{2} |60 + 80 - 12| = \frac{1}{2}(128)$$

Step 3. The lot covers

$$\boxed{64 \text{ m}^2}$$

Common wrong answer: 80, the smallest rectangle that surrounds the lot (10 wide by 8 tall). The lot is strictly inside that rectangle, so the bounding box is an upper bound, never the answer.

8. Is $W(-2, 1)$, $X(3, 3)$, $Y(5, -2)$, $Z(0, -4)$ a square?

Step 1. A square needs two separate facts: four congruent sides, and one right angle. Congruence alone leaves a rhombus, so distances settle half the claim and slopes settle the other half.

Step 2. $WX = \sqrt{(3 - (-2))^2 + (3 - 1)^2} = \sqrt{25 + 4} = \boxed{\sqrt{29}}$

The other three sides give the same radical: $XY = \sqrt{4 + 25}$, $YZ = \sqrt{25 + 4}$, $ZW = \sqrt{4 + 25}$.

Step 3. Shoelace on the four vertices in order gives $\frac{1}{2} |-2(3 - (-4)) + 3(-2 - 1) + 5(-4 - 3) + 0(1 - (-2))| = \frac{1}{2}(58)$, so the area is

$$\boxed{29 \text{ square units}}$$

Written justification (manual review). A complete answer states both halves: (i) all four sides measure $\sqrt{29}$, so the figure is at least a rhombus; (ii) $\text{slope}(WX) = \frac{2}{5}$ and $\text{slope}(XY) = -\frac{5}{2}$, whose product is -1 , so the angle at X is right — and a rhombus with one right angle has four. The designer is correct. Accept any write-up that establishes congruent sides AND perpendicularity; a write-up that checks only the four lengths has proved a rhombus, not a square, and should be sent back for the slope step.

Consistency note: the area equals $(\sqrt{29})^2 = 29$, which is what a square's area must do. That agreement is itself evidence for the claim.